

FAIR-FRFT: Fairness-Aware Fractional Fourier Transform for Deepfake Detection

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Recent advances in deepfake detection have revealed significant fairness challenges, as many existing models exhibit biased performance across demographic groups. Such disparities pose critical risks to AI safety and the ethical deployment of forensic technologies. In this work, we present **FAIR-FRFT**, a novel framework that enforces demographic invariance through reverse gradient learning. Unlike prior approaches that incorporate fairness constraints into existing architectures, FAIR-FRFT adopts a fundamentally different design based on a Fractional Fourier Transform (FrFT)-driven adversarial autoencoder. Operating in the hybrid spatial-frequency domain, the proposed model learns latent representations that preserve semantic deepfake cues while suppressing demographic attributes such as skin tone and facial structure. This formulation enhances the model’s ability to capture frequency-domain artifacts that are less correlated with demographic information. Extensive experiments demonstrate that FAIR-FRFT achieves robust and demographically fair deepfake detection, ensuring balanced performance across diverse populations. The code for FAIR-FRFT is available at: <https://github.com/iMmOrTaL2121/FAIR-FRFT-Fairness-Aware-Fractional-Fourier-Transform-for-Deepfake-Detection>

Introduction

Deepfake detectors often achieve high overall accuracy but show large performance gaps across demographic subgroups - for example, higher false positive rates for darker skin tones. Such disparities arise from biased training data and unintended correlations between appearance features and detection performance.

Traditional approaches to fairness in deepfake detection have focused on in-processing (as in Zafar paper ref) bias mitigation through loss function modifications. Pre-processing techniques like data augmentation might help but upto a very limited extent. However, these methods treat fairness as an afterthought rather than a fundamental design principle. We argue that truly fair deepfake detection requires learning representations that are inherently demographic-agnostic from the ground up. Our key insight

is that fairness should be achieved through architectural design rather than constraint optimization, specifically by learning latent representations that focus on semantic content rather than superficial appearance variations (skin tone, facial structure) that correlate with demographics, preserving only deepfake-indicative structural patterns through adversarial training with gradient reversal.

Furthermore, we leverage the observation that deepfake artifacts are more evident in the frequency domain than in the spatial domain (Liu et al. 2021). Deepfake generation processes introduce subtle frequency distortions that are less correlated with demographic features, making frequency-domain analysis particularly suitable for fair detection. By operating in the hybrid spatial-frequency domain using Fractional Fourier Transform, our method can better capture these discriminative artifacts while naturally reducing demographic bias.

Related Work

Fairness in machine learning has been widely explored across domains (Mehrabi et al. 2021; Barocas, Hardt, and Narayanan 2019), though most studies focus on tabular classification tasks. Fairness in computer vision, particularly involving complex visual features, remains underexplored. Recent advances in disentanglement learning for fair representation learning have been mentioned in the paper (Pfau et al. 2020).

Deepfake detection has evolved from early CNN-based architectures (V and Joy 2023) to hybrid spatial-frequency methods. MesoNet (Afchar et al. 2018) is designed as a compact convolutional neural network for detecting facial video forgery, while SRM-based methods (Fridrich & Kodovský, 2013) exploited frequency-domain statistical cues. Spatial-phase learning (Liu et al., 2021) further demonstrated the benefits of frequency information. However, most of these models neglect demographic fairness, leading to unequal performance across gender, age, and ethnicity.

Recent studies highlight fairness gaps in deepfake detection (Lin et al., 2024a). DAW-FDD (Demographic-Aware Weighted Fair Deepfake Detection) (Ju et al. 2020) introduced an in-processing fairness framework using demographic-aware loss weighting to balance group performance. While effective within domains, DAW-FDD struggles to generalize fairness cross-domain. DAG-FDD

(Demographic-Agnostic Fair Deepfake Detection) (Ju et al. 2020) addressed this by disentangling demographic and forgery features to enhance fairness transfer. Yet, it remains constraint-based, focusing on fairness regularization rather than intrinsically fair representations. PG-FDD (Privacy-Guided Fair Deepfake Detection) extends this paradigm by jointly constraining demographic information leakage for privacy and fairness, but still operates within the same constraint-driven framework.

Collectively, these approaches reveal that fairness and generalization are deeply interlinked in deepfake detection. While DAW-FDD, DAG-FDD, and PG-FDD represent key milestones, they underscore the need for fundamentally fair, demographically agnostic representation learning—an objective our proposed method advances.

Our Contribution

Unlike existing approaches that treat fairness as a constraint, our FAIR-FRFT method fundamentally rethinks fairness in deepfake detection by designing an adversarial autoencoder architecture that inherently learns demographic-agnostic representations through gradient reversal. Our method removes demographic information from latent representations using adversarial training, ensuring that learned features cannot be used to predict demographic attributes. We leverage the unique properties of Fractional Fourier Transform to capture deepfake artifacts that are less correlated with demographic features, providing a natural foundation for fair representation learning.

Background

We consider a face image dataset $x \in \mathbb{R}^{H \times W \times C}$ with labels $y \in \{0, 1\}$ (real/fake) and demographic group $g \in \mathcal{G}$. The detector outputs a score $s \in \mathbb{R}$ and must (i) distinguish real from fake, (ii) reduce demographic bias, and (iii) exploit both spatial and frequency cues (Liu et al. 2021).

Fractional Fourier Transform Processing

The key innovation of our approach lies in leveraging the Fractional Fourier Transform (FrFT) to capture continuous hybrid spatial-frequency information that standard Fourier analysis misses and are more discriminative for fake detection. This transformation is defined as $x' = F_\alpha^{(d)}(x)$, where $\alpha \in [0, 1]$ controls the interpolation between spatial and frequency domains.

The Fractional Fourier Transform is mathematically expressed as:

$$F_\alpha[f](u) = \int_{-\infty}^{\infty} f(t) K_\alpha(t, u) dt$$

where $K_\alpha(t, u) = A_\alpha \exp(i\pi(t^2 \cot \alpha - 2tu \csc \alpha + u^2 \cot \alpha))$ and $A_\alpha = \sqrt{1 - i \cot \alpha}$ is the normalization constant.

Adversarial Autoencoder Architecture

The FrFT-transformed image x' is processed through our adversarial autoencoder pipeline. The encoder network E_θ :

$\mathbb{R}^{H \times W \times C} \rightarrow \mathbb{R}^d$ maps the FrFT-transformed image to a d -dimensional latent representation $z = E_\theta(x')$. The decoder network $G_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^{H \times W \times C}$ reconstructs the spatial-domain image as $\hat{x} = G_\theta(z)$. To enforce consistency, we re-encode the reconstructed image through FrFT: $\hat{x}' = F_\alpha^{(d)}(\hat{x})$ and $\hat{z} = E_\theta(\hat{x}')$.

Latent Consistency Objective

Instead of pixel reconstruction, we enforce latent consistency to learn semantically meaningful representations while avoiding demographic bias:

$$\mathcal{L}_{\text{lat}} = \|\hat{z} - \text{sg}(z)\|_2^2 + \beta \cdot (1 - \cos(\hat{z}, \text{sg}(z)))$$

where $\text{sg}(\cdot)$ denotes the stop-gradient operator. This cycle-consistency forces the autoencoder to learn semantically meaningful representations while ignoring superficial appearance variations (skin tone, facial structure) that correlate with demographics, preserving only deepfake-indicative structural patterns. This fundamental insight motivates our autoencoder-based approach: by focusing on semantic content rather than pixel-level details, we naturally avoid demographic bias while maintaining detection performance.

Adversarial Demographic Debiasing

Building on the latent consistency foundation, our core innovation is the use of gradient reversal to actively remove demographic information from latent representations. Building on recent advances in disentanglement learning by (Pfau et al. 2020), we train a demographic classifier $D_\phi : \mathbb{R}^d \rightarrow \mathbb{R}^{|\mathcal{G}|}$ that predicts demographic group g from the encoded features:

$$\mathcal{L}_{\text{adv-clf}} = \mathbb{E}_{x, g} [\text{CE}(D_\phi(\text{GRL}(z)), g)],$$

where $\text{GRL}(\cdot)$ applies gradient reversal to the latent representation z .

The key insight is that when D_ϕ successfully classifies demographics, we want to penalize the encoder E_θ for producing representations that contain demographic information. This is achieved through gradient reversal:

$$\nabla_\theta \mathcal{L}_{\text{AE}} = \nabla_\theta \mathcal{L}_{\text{lat}} + \lambda_{\text{adv}} \nabla_\theta (-\mathcal{L}_{\text{adv-clf}})$$

where the negative sign in front of $\mathcal{L}_{\text{adv-clf}}$ implements gradient reversal. The gradient reversal layer (GRL) is defined as:

$$\text{GRL}(x) = x \quad , \quad \frac{\partial \text{GRL}(x)}{\partial x} = -\lambda_{\text{grl}}$$

where λ_{grl} is the gradient reversal strength. This creates a minimax game where the encoder learns to produce representations that are both deepfake-discriminative and demographic-agnostic. The adversarial training objective can be formulated as:

$$\min_\theta \max_\phi \mathcal{L}_{\text{AE}} = \mathcal{L}_{\text{lat}}(\theta) + \lambda_{\text{adv}} \mathcal{L}_{\text{adv-clf}}(\theta, \phi)$$

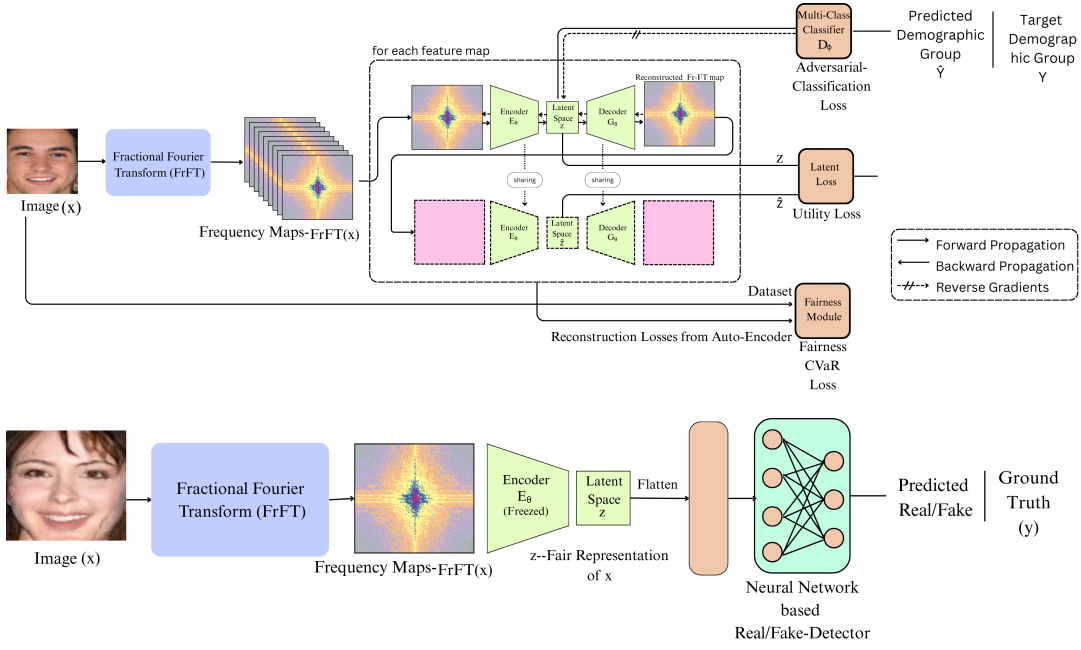


Figure 1: FAIR-FRFT Architecture. Top: FrFT-transformed input x' is encoded by E_θ into latent z , reconstructed by G_θ to $\hat{x}$, and re-encoded for consistency. The model includes demographic adversary D_ϕ , real/fake head R_ψ , and CVaR fairness loss.

CVaR Fairness

To protect vulnerable subgroups Conditional Value at Risk (CVaR) is used:

$$\mathcal{L}_{\text{fair}} = \max_{y,a} \text{CVaR}_\alpha(\ell(x,y) | y, a),$$

which focuses on worst-case performance across all demographic-label combinations, ensuring Equalized Odds.

Training Objective

The autoencoder training minimizes:

$$\min_{\theta} \max_{\phi} \lambda_{\text{lat}} \mathcal{L}_{\text{lat}} + \lambda_{\text{fair}} \mathcal{L}_{\text{fair}} - \lambda_{\text{adv}} \mathcal{L}_{\text{adv-clf}}.$$

$$\mathcal{L}_{\text{lat}} = \|\hat{z} - \text{sg}(z)\|_2^2 + \beta \cdot (1 - \cos(\hat{z}, \text{sg}(z))), \quad \beta = 1.0$$

$$\mathcal{L}_{\text{adv-clf}} = \mathbb{E}_{x,g} [\text{CE}(D_\phi(\text{GRL}(z)), g)]$$

$$\mathcal{L}_{\text{fair}} = \max_{y,a} \text{CVaR}_\alpha(\ell(x,y) | y, a),$$

After convergence, the encoder is frozen and a separate classifier C_ψ is trained on z to detect real/fake with standard cross-entropy:

$$\mathcal{L}_{\text{clf}} = \mathbb{E}_{x,y} [\text{CE}(C_\psi(z), y)].$$

The detailed algorithm is mentioned in the Appendix .

Training Strategy

The training process follows a two-stage optimization strategy that aligns with our architectural design:

Stage 1 - Fair Representation Learning: We first optimize the adversarial autoencoder to learn demographic-agnostic representations:

$$\begin{aligned} \theta^{(1)}, \phi^* &= \arg \min_{\theta} \max_{\phi} \mathcal{L}_{\text{Total}}(\theta, \phi) \\ &= \lambda_{\text{lat}} \mathcal{L}_{\text{lat}}(\theta) + \lambda_{\text{fair}} \mathcal{L}_{\text{fair}}(\theta) \\ &\quad - \lambda_{\text{adv}} \mathcal{L}_{\text{adv-clf}}(\theta, \phi) \end{aligned} \quad (1)$$

This stage learns latent representations z that are both semantically consistent and demographic-blind through the minimax game between the encoder E_θ and the demographic classifier D_ϕ .

Stage 2 - Binary Classification: After the encoder converges, we freeze E_θ and train the real/fake classifier:

$$\psi^* = \arg \min_{\psi} \mathcal{L}_{\text{clf}}(\psi; \theta^{(1)}) = \mathbb{E}_{x,y} [\text{CE}(C_\psi(z), y)]$$

Our implementation employs several key training techniques:

1. **Fairness Warmup:** To prevent early convergence to biased solutions, we gradually introduce the fairness loss:

$$\lambda_{\text{fair}}(t) = \begin{cases} 0 & \text{if } t < 3 \\ \lambda_{\text{fair}}^{\text{max}} \cdot \frac{t-3}{\text{total epochs}-3} & \text{otherwise} \end{cases}$$

This allows the autoencoder to first learn meaningful representations before enforcing demographic invariance.

2. **Adaptive Gradient Reversal:** The gradient reversal strength increases progressively:

$$\lambda_{\text{grl}}(t) = \min(0.5, t/10.0)$$

ensuring smooth transition from representation learning to demographic debiasing.

3. **Early Stopping:** Dual-criteria monitoring based on both latent consistency and fairness:

$$\text{improved} = (\mathcal{L}_{\text{lat}} < 0.995 \cdot \mathcal{L}_{\text{lat}}^{\text{best}}) \vee (\mathcal{L}_{\text{fair}} < 0.995 \cdot \mathcal{L}_{\text{fair}}^{\text{best}})$$

with patience of 7 epochs, preventing overfitting while maintaining fairness.

4. **Gradient Clipping:** Global norm clipping at 5.0 stabilizes adversarial training dynamics.
5. **Class Weighting:** Automatic positive class weight computation handles real/fake data imbalance during Stage 2.

The complete training runs for 30 epochs in Stage 1 (autoencoder) and 50 epochs in Stage 2 (classifier), with distributed training across multiple GPUs and mixed precision optimization for computational efficiency.

Experimental Setup

Datasets

We evaluate our method on the Celeb-DF dataset, a million-scale, demographically annotated benchmark comprising real faces, deepfake frames, and GAN- or diffusion-generated images. It offers balanced distributions across gender, age, and skin tone, with high-quality annotations verified through automated and manual analysis for robust fairness evaluation.

Implementation Details

Our implementation uses TensorFlow with distributed training and mixed precision optimization:

Architecture Specifications: Our encoder employs a 3-layer CNN (32-64-128 filters) with no activation, Layer-Normalization, and Gaussian noise (std=0.05). The decoder is symmetric with upsampling layers and skip connections. The demographic classifier uses a 2-layer MLP with Layer-Normalization and MaxNorm constraints (max norm=3.0), while the real/fake head employs a 2-layer MLP with Layer-Normalization and Dropout (0.2). We use a latent dimension of 128 optimized for memory efficiency. The FrFT implementation uses 2D FrFT with order=3 via chirp-FFT-chirp approach for GPU optimization, and images are processed at 128x128x1 (grayscale) for computational efficiency.

Training Configuration: We use AdamW optimizer with learning rate 1e-4 and weight decay 1e-4, with batch size 64 and distributed training across 2 GPUs using MirroredStrategy. Mixed precision (float16) is employed for improved performance and memory usage. Early stopping uses patience of 7 epochs based on latent consistency and fairness metrics, with fairness warmup over 3 epochs for gradual introduction of fairness loss. Global norm clipping at 5.0 ensures training stability. The total training involves 30 epochs for the autoencoder and 50 epochs for the binary classifier.

Baseline Methods

We compare against the following state-of-the-art approaches: Xception, Xception with PG-FDD, Xception with DAW-FDD, Xception with DAG-FDD, and SRM (frequency-based method).

Results

Detection Performance

Table 1 presents a comprehensive comparison of FAIR-FRFT against state-of-the-art deepfake detection methods on the Celeb-DF dataset. Our approach achieves competitive detection performance while maintaining strong fairness guarantees in multiple metrics. We have compared AUC as utility metric and four fairness metrics as shown in the results table in the Appendix.

Future Work

Promising directions include extending the approach to multiple demographic attributes simultaneously, learning optimal FrFT parameters for different deepfake artifacts, and optimizing for real-time deployment. Addressing individual fairness in deepfake detection presents a particularly challenging direction (Hou et al. 2025), as the fundamental contradiction between individual fairness principles and the nature of deepfake detection tasks requires novel approaches that can handle similar images belonging to opposite classes.

Conclusion

We present FAIR-FRFT, a novel fairness-aware deepfake detection framework that leverages Fractional Fourier Transform to capture spatially-localized frequency artifacts while enforcing demographic parity through adversarial training and CVaR-based tail risk minimization. Unlike existing approaches that treat fairness as a constraint, our method employs gradient reversal to actively learn demographic-agnostic latent representations from the ground up.

Our experimental results demonstrate that FAIR-FRFT achieves competitive performance (AUC of 0.926, CVAR of 0.947) while significantly reducing bias, showing the effectiveness of treating fairness as a fundamental design principle. This work opens new avenues for fairness-aware computer vision and demonstrates the potential of hybrid spatial-frequency domain analysis for fair representation learning.

Appendix

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Algorithm 1: Fairness-Aware Adversarial Autoencoder Training

```
1: Input: Dataset  $\{(x_i, y_i, g_i)\}$ , hyperparameters  $\lambda_{\text{lat}}, \lambda_{\text{fair}}, \lambda_{\text{adv}}$ 
2: Initialize: Encoder  $E_\theta$ , Decoder  $G_\theta$ , Demographic Classifier  $D_\phi$ 
3: Stage 1 – Fair Representation Learning:
4: for epoch = 1 to 30 do
5:   for each batch  $\{(x, y, g)\}$  do
6:     FrFT transform:  $x' \leftarrow F_\alpha^{(d)}(x)$ 
7:     Encode:  $z \leftarrow E_\theta(x')$ 
8:     Decode and cycle:  $\hat{x} \leftarrow G_\theta(z), \hat{x}' \leftarrow F_\alpha^{(d)}(\hat{x}), \hat{z} \leftarrow E_\theta(\hat{x}')$ 
9:     Compute latent consistency:

$$\mathcal{L}_{\text{lat}} = \|\hat{z} - \text{sg}(z)\|_2^2 + \beta(1 - \cos(\hat{z}, \text{sg}(z)))$$

10:    Compute adversarial loss:

$$\mathcal{L}_{\text{adv-clf}} = \text{CE}(D_\phi(\text{GRL}(z)), g)$$

11:    Compute CVaR fairness loss:

$$\mathcal{L}_{\text{fair}} = \max_{y, a} \text{CVaR}_\alpha(\ell(x, y) \mid y, a)$$

12:    Apply warmup schedule:

$$\lambda_{\text{fair}}(t) \leftarrow \max\left(0, \lambda_{\text{fair}}^{\text{max}} \cdot \frac{t-3}{27}\right)$$

13:    Update model parameters:

$$\mathcal{L}_{\text{AE}} = \lambda_{\text{lat}}\mathcal{L}_{\text{lat}} + \lambda_{\text{fair}}\mathcal{L}_{\text{fair}} - \lambda_{\text{adv}}\mathcal{L}_{\text{adv-clf}}$$


$$\theta, \phi \leftarrow \text{AdamW}(\mathcal{L}_{\text{AE}})$$

14:   end for
15:   Check early stopping based on improvement in  $\mathcal{L}_{\text{lat}}$  and  $\mathcal{L}_{\text{fair}}$ 
16: end for
17: Stage 2 – Binary Classification:
18: Freeze encoder  $E_\theta$ 
19: for epoch = 1 to 50 do
20:   for each batch  $\{(x, y)\}$  do
21:     Extract features:  $z \leftarrow E_\theta(F_\alpha^{(d)}(x))$ 
22:     Train classifier:

$$\mathcal{L}_{\text{clf}} = \text{CE}(C_\psi(z), y)$$


$$\psi \leftarrow \text{AdamW}(\mathcal{L}_{\text{clf}})$$

23:   end for
24:   Check early stopping based on validation AUC
25: end for
```

Table 1: Detection Performance Comparison on AI-Face Dataset

Method	AUC	FPR	MEO	DP	OAE
XceptionNet	0.9580	0.1432	0.1845	0.2597	0.3148
XNet (with PG-FDD)	0.8749	0.0128	0.0345	0.0497	0.2883
XNet (with DAW-FDD)	0.9567	0.1286	0.1043	0.1427	0.2764
XNet (with DAG-FDD)	0.9506	0.0052	0.0824	0.1247	0.2794
FAIR-FRFT	0.9543	0.0348	0.0949	0.2454	0.0688

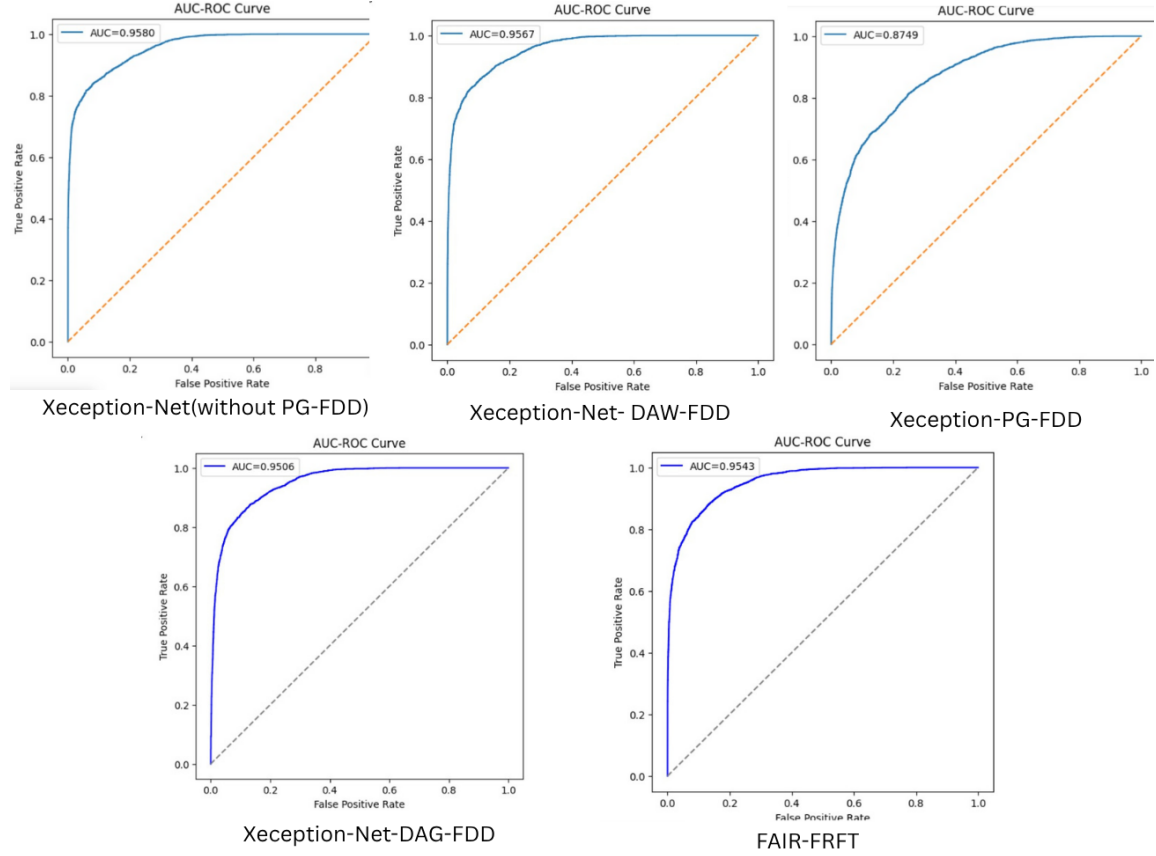


Figure 2: Performance metrics showing AUC.

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